

The Triadic Measurement Substrate

$$\bar{B} = 3\bar{\alpha}$$

Volume 2

Paper 1: The Now-Surface: Gravity, Black Holes, and Dynamic Dark Energy from Triadic Confirmation

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Paper 1

The Now-Surface: Gravity, Black Holes, and Dynamic Dark Energy from Triadic Confirmation

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Status of Results

This paper formalizes a single intensive derivation campaign. It distinguishes six categories of claim, each labeled where it appears in the text. The categories continue the labeling conventions established in Volume 1, extended with two categories appropriate to this paper's predictive and coefficient-resolving content. No claim is upgraded beyond what its derivation earns; quantities matched to observation are marked Consistent, not Derived, wherever they depend on the empirical level identity k or on contested observational inputs.

Derived. Results that follow from the five axioms, the Principle of Generative Economy, and U_{sh} by explicit construction or proof, or from results already derived in Volume 1 by explicit inheritance. The two zeros of the structured-novelty function and the black-hole/heat-death identification (§ 2.1); the horizon as a two-dimensional agent manifold with its non-conditional area law (§ 2.2); the confirmation (Unruh-form) temperature with exact $1/2\pi$ coefficient (§ 3.1); exact Lorentz invariance at second order in wavevector by direct Fourier audit of the FCC Laplacian (§ 3.2); the horizon's unavoidable radiation from U_{sh} fluctuations across the causal seal (§ 4.1); the reinterpretation of infalling information into Ξ and the resulting perspective boundary (§ 4.3); the within/cross weight $\sqrt{2}$ from S_3 representation theory (§ 5.1); the triadic root $|S_3| = 6$ of the level-advance (§ 5.3); and dark energy's dynamic necessity, with $w = -1$ structurally excluded (§ 7), are derived in this sense.

Form-derived. Results whose structure is derived but whose single remaining coefficient is gated on the within/cross coupling identity of § 5. The Schwarzschild solution and its coincidence of the general-relativistic horizon with the $\hat{R}$ surface (§ 2.3); the Einstein field equations via the Jacobson construction on the derived area law, temperature, and second-order isotropy, with the Poisson structure as the Newtonian-limit shadow (§ 3.3); and the Hawking temperature with its parameter-free temperature–mass relation (§ 4.2) are form-derived in this sense.

Closed. Quantities Volume 1 flagged open, here reduced to the triadic alphabet and grammar and resolved within this paper. The black-hole coefficient $4/\sqrt{3}$ as a count over a triangular measure, and the gravitational ratio $\sqrt{6}/9$ as within/cross weight times triangular measure times triangle count (§ 5.2), are closed in this sense.

Predictions. Falsifiable claims, parameter-free in form, that distinguish TMS from standard theory but whose magnitudes are not closed here. The equivalence principle as a depth-suppression theorem, with

the suppression sign derived and the magnitude a bound (§ 6); and the forced crossing of $w = -1$ by the dark-energy equation of state (§ 7), are predictions in this sense.

Consistent. Quantities that match observation given our universe’s inputs, not parameter-free. The numerical dark-energy equation-of-state parameters and crossing redshift (§ 7); and the level identity $k \approx 79$, triple-consistent across the cosmological-constant ratio, the equivalence-principle bound, and the dark-energy evolution but order-of-magnitude and resting on Volume 1’s steady-state assumption (§ 8), are consistent in this sense.

Deferred / Open. Items flagged explicitly and not closed in this paper. The literal sequential-sampling realization of $|S_3| = 6$ (§ 5.3); the numerical equation-of-state parameters from $k(a)$ (§ 7); and a formal, non-steady-state derivation of $k \approx 79$ (§ 8) remain open. The area-coupling factor of two ($= 2$) and the geometric extent overhead area-coupling factor of two ($= 2$) and the geometric extent overhead $\epsilon^{(k)}$ ($= 0$) are derived in § 5.5.

This six-way labeling applies throughout the paper. Every claim in the body text falls into exactly one category, and every category’s status is visible at the point of claim.

Abstract. *This paper introduces the now-surface — the two-dimensional agent manifold on which confirmation is actively occurring, as distinct from the three-dimensional record it leaves behind — and shows that this single structural object derives the central phenomena of the material universe that Volume 1 left form-complete but coefficient-open. A black hole is derived as a now-surface that reaches the full-novelty zero and folds off via the reinterpretation operator $\hat{R}$; its horizon is derived as a two-dimensional agent manifold; the Schwarzschild metric follows as the vacuum solution of Einstein equations obtained, in the manner of Jacobson, from a derived area law and a re-derived Unruh temperature, resting on Lorentz invariance shown exact at second order in wavevector by direct audit of the FCC lattice. The open gravitational coupling reduces to a single triadic identity, reframed as one alphabet of three letters with a derived grammar that spells the black-hole coefficient and the gravitational coupling and largely resolves the coupling-universality keystone of the substrate stabilizer code. Dark energy is derived to be dynamic by necessity — a residual on a perpetually advancing now-surface cannot hold constant density — and its equation of state is predicted to cross the cosmological-constant value, a parameter-free distinction from the standard model that qualitatively matches recent survey indications of evolving dark energy. Finally, the framework’s single empirical input, the universe’s hierarchical depth, is pinned to approximately seventy-nine from the cosmological-constant ratio and shown consistent across the equivalence-principle bound and the dark-energy evolution. The material universe reduces to one geometric primitive, one axiom, and one integer.*

1. The Now-Surface

A confirmation event is active only at the frontier where agents are presently locking bits; everything behind that frontier is settled record. We call this active frontier the now-surface: the two-dimensional agent manifold on which confirmation is currently occurring, as distinct from the three-dimensional causal record, or wake, that accumulates behind it. The distinction is structural and follows from the substrate of Volume 1: confirmation is a local locking event, so at any flop only a codimension-one frontier is active while the locked interior persists.

The reinterpretation operator $\hat{R}$ (Volume 1, Definition 0.0.67) fires when a region’s structured novelty $v = (1 - \rho) \cdot H(\rho)$ reaches zero. This paper establishes that the now-surface, taken purely as this structural object, derives the central material phenomena Volume 1 left open: black holes, the metric, gravity’s coupling, and dark energy. The same construct will carry the analysis of bounded observers and graded subjecthood

in the papers that follow; here it is developed entirely on the material side, and no property of observers is assumed.

2. The Black Hole

2.1 Two zeros of one function

Structured novelty $v = (1 - \rho)H(\rho)$ vanishes at both $\rho \rightarrow 0$ (noise death) and $\rho \rightarrow 1$ (saturation). Both factors are derived: $(1 - \rho)$ from U_{sh} maximum-entropy sampling, and $H(\rho)$ from binary-bit source coding. A black hole and heat death are the same structural transition approached from opposite ends; both fold off a daughter universe via $\hat{R}$.

2.2 The horizon as a two-dimensional agent manifold

Volume 1 (p. 105) proves that the identification “confirmed bit becomes new agent” is forced by the axioms. The now-surface/wake split forces which bits constitute the daughter manifold: the two-dimensional now-surface frontier at saturation, not the three-dimensional wake. Axiom scale-independence forces this manifold to be two-dimensional. The horizon is therefore necessarily a two-dimensional agent manifold, and the area law sits on it non-conditionally, in the sense in which horizon entropy scales with area (Bekenstein, 1973).

2.3 The Schwarzschild solution

Solving the vacuum field equations under static spherical symmetry yields $f(r) = 1 - r_s/r$ with $r_s = 2GM/c^2$. The general-relativistic horizon ($f = 0$) coincides with the $\hat{R}$ surface (exterior confirmation throughput vanishing), via the identification of gravitational time dilation $\sqrt{f}$ with the local flop-rate ratio. This retro-explains the failure of a naive discrete-level redshift route: the continuum metric gives a finite r_s , whereas discrete level-stacking piles redshift at infinite radius.

3. The Metric, on Verified Lorentz Invariance

3.1 Confirmation temperature

Re-derived from U_{sh} white noise along a uniformly accelerated worldline: the proper-time correlation is periodic in imaginary time with period $2\pi c/a$, giving $T = \hbar a / (2\pi c k_B)$, the Unruh form (Unruh, 1976) with coefficient $1/2\pi$ exact.

3.2 Exact Lorentz invariance at second order

Direct Fourier audit of the FCC Laplacian: angular anisotropy scales as the fourth power of wavevector magnitude. The second-order term — the wave equation, the order at which the metric lives — is exactly isotropic and hence exactly Lorentz invariant. The first anisotropy enters at fourth order, Planck-suppressed: a falsifiable signature, not a defect.

3.3 The Einstein equations and the Poisson structure

The Einstein field equations follow from three results already established, combined by the Jacobson construction (Jacobson, 1995). The inherited pieces are named first. (i) The area law of § 2.2: entropy is proportional to the two-dimensional now-surface area, derived non-conditionally there because the horizon

is forced to be a two-dimensional agent manifold. (ii) The confirmation temperature of § 3.1: a uniformly accelerated worldline through U_{sh} sees a thermal spectrum at $T = \hbar a / (2\pi c k_B)$, with the $1/2\pi$ coefficient exact. (iii) The exact second-order Lorentz invariance of § 3.2: the wave-equation order is exactly isotropic by direct Fourier audit of the FCC Laplacian, so the local Rindler structure the construction requires is available without approximation.

Given these three, the Jacobson construction is forced rather than imported. Demanding the thermodynamic relation $\delta Q = T dS$ hold across every local causal horizon — with δQ the confirmation-flux across the now-surface and T the § 3.1 temperature — fixes the relation between horizon-area variation and the energy flux through it. Because the area law (i) supplies the entropy, the temperature (ii) supplies T , and the second-order isotropy (iii) guarantees the local Rindler horizons are exactly those of a Lorentz-invariant continuum, the only field equation consistent with $\delta Q = T dS$ holding at every point is the Einstein equation. Schwarzschild is its vacuum solution under static spherical symmetry, recovering § 2.3.

The same picture yields the Newtonian limit directly, with no further assumption. Read gravity as the resolution degradation of confirmation as the now-surface approaches its information ceiling: channel-fullness $\rho(r) \sim 1/r^2$ plays the role of the gravitational field, and the resolution deficit $\Phi \sim 1/r$ plays the role of the potential, so that $\nabla^2 \Phi = \text{source}$ is the Poisson equation. Entropy is primary and embedding depth is its integral; the metric depth is therefore emergent, not fundamental. This is the continuum-limit shadow of the same construction, and it carries no open coefficient of its own: the area-coupling factor of two that § 5 once left open is now derived (§ 5.5). This route places TMS within the thermodynamic and entropic-gravity program — the derivation of the field equations as an equation of state (Jacobson, 1995) and the reading of gravity as an entropic force (Verlinde, 2011) — from which it differs in supplying the underlying microstructure: the now-surface and its confirmation flux, rather than an unspecified holographic screen, carry the entropy.

4. Hawking Radiation and the Perspective Cut

4.1 The horizon must radiate

U_{sh} fluctuates at every agent, the horizon included. A horizon agent straddles the causal seal — its interior side faces the Ξ -wake (sealed daughter), its exterior side our now-surface. A fluctuation fires a confirmation with one element sealed in Ξ and its partner radiating outward: substrate horizon pair-production. The field never rests, so the emission is unavoidable.

4.2 The thermal spectrum

A static observer near r_s is accelerated; U_{sh} along that worldline is thermal (§ 3.1). At the horizon the acceleration is the surface gravity $\kappa = c^4 / (4GM)$, giving $T_H = \hbar \kappa / (2\pi c k_B) = \hbar c^3 / (8\pi GM k_B)$, the Hawking temperature: a solar-mass hole gives 6.18×10^{-8} K (textbook 6.2×10^{-8}). Smaller holes are hotter; the hole evaporates. The $1/2\pi$ coefficient and the temperature–mass relation are parameter-free; the absolute scale inherits the coupling identity of § 5 and introduces no separate open problem.

4.3 Information and the perspective boundary

The infalling information is not destroyed: the sealed element joins the Ξ -wake as the daughter substrate's initial data (Volume 1, p. 109); the partner radiates. Information is reinterpreted into Ξ , not erased — the paradox dissolves. This reinterpretation across a causal seal is a structural perspective boundary: a partition

of confirmation into what is sealed within one frame and what is emitted to others. The construct is recorded here as a derived feature of the material substrate; the analysis of bounded observers that builds on it is the subject of the papers that follow. Hawking radiation is, in this sense, the material-side instance of a perspective boundary.

5. The Coupling: One Alphabet, One Grammar

Four area-based routes to the gravitational coupling unify to $\sqrt{3}/4$ per bit, modulo forced factors (4π from Gauss's law, $\ln 2$ from the bit-nat conversion, and a factor of two now derived in § 5.5). The sole standout is the cross-level structural coupling $\sqrt{2}/12$; the ratio of the two is $\sqrt{6}/9$.

5.1 The within/cross weight

From the S_3 representation theory of the triadic event: the surviving trivial modes — the bit channel T (three diagonal components, normalization $1/\sqrt{3}$) and the relation channel R (six off-diagonal components, normalization $1/\sqrt{6}$) — have normalization ratio $\sqrt{2}$, the same $\sqrt{2}$ as the FCC face-diagonal. Algebra and geometry converge.

5.2 The alphabet and the grammar

The relevant constants form a three-letter alphabet read dimensionally: 3 is the triangle (two-dimensional minimum confirmation, $\bar{B} = 3\bar{\alpha}$); 4 is the tetrahedron (three-dimensional minimum, $\bar{B} = 4\bar{\alpha}^3$); $\sqrt{2}$ is the within/cross weight. The grammar, derived from the clean black-hole coefficient with no target value in view, is: counts enter linearly; a two-dimensional triangular measure carries $\sqrt{3}$; the within/cross weight carries $\sqrt{2}$; depth carries $(1/\sqrt{2})^k$.

The black-hole coefficient $4/\sqrt{3}$ spells as a count over a triangular measure. The gravitational ratio $\sqrt{6}/9$ spells as $\sqrt{2} \cdot (1/\sqrt{3}) \cdot (1/3)$ — within/cross weight, triangular measure, and triangle count, each in its assigned role.

5.3 The triadic root of the level-advance

The level-advance time-ratio is forced to $|S_3| = 6$: meta-confirmation of three subsystems must be role-independent, hence samples all six orderings; and it must be the full symmetric group, not merely the cyclic subgroup, because Volume 1's own seven-of-nine vacuum-mode cancellation kills the sign mode, which requires the odd permutations. The “3” in the length-ratio $3\sqrt{2}$ is therefore triadic, the gravitational grammar holds unconditionally, and this $|S_3| = 6$ converges with the W-point hypothesis ($24 = 4 \times 6$). The literal sequential-sampling realization remains open.

5.4 The coupling-universality keystone

Volume 1's five bit-footprint coupling coefficients are the alphabet in scale-invariant roles: $\alpha_{\text{surround}} = \sqrt{2}$, $\alpha_{\text{subsys}} = 3$, $\alpha_d = \alpha_{\text{local}} = 1$, and $\alpha_k = (1/\sqrt{2})^k$. Four of the five are grammar-forced universal; the fifth carries the derived depth-dilution, with no residual. The keystone Volume 1 flagged as open is thereby resolved completely: the per-level structural overhead δ_k that earlier drafts carried as an undetermined factor in α_k is now derived to be identically one. The content of this is not the trivial observation that a ratio of identical quantities is one; it is that the meta-tetrahedral geometry *is* identical at every level, which is a computed and non-trivial result — the form-invariance fixed point (§ 4.7; Paper 3) establishes by explicit

enumeration that the coordination number, triangle count, and tetrahedron count per site (twelve, twenty-four, eight) recur unchanged across levels, a fact that could have failed and does not. The question δ_k answered was whether the depth-coupling carries any per-level overhead *beyond* the speed-dilution $(1/\sqrt{2})^k$; since the propagation speed already absorbs the dynamical (flop-time) cost through $c^{(k)} = (1/\sqrt{2})^k$, and the fixed point shows the residual geometric content is level-invariant, no third overhead remains. Two points forestall the natural objection that the recursion hides a correction. First, the level-invariance is not assumed in the induction but *counted*: the meta-lattice coordination, triangle, and tetrahedron numbers are obtained by direct enumeration of the neighbours of a site built from the level below, and that count makes no reference to the coupling α_k or to δ_k , so $\delta_k = 1$ is read off an independently established invariance rather than presupposed to establish it. Second, δ_k is the *bulk* coupling — α_k governs propagation through the repeating lattice — and finite-patch boundary effects (fewer neighbours at a surface site, stabilizer leakage at an edge) are a separate correction of order surface-to-volume, $O(1/L)$ in patch size L , which vanishes in the bulk limit and does not enter the bulk coupling. Given that computed and boundary-independent invariance, $\delta_k = 1$ and $\alpha_k = (1/\sqrt{2})^k$ with no open residual; the depth-coupling is fully determined.

5.5 Two constants closed: the area-coupling factor of two and the extent overhead

Two coefficients that Volume 1 carried as open close exactly, each once its correct dimensional home is identified.

The area-coupling factor of two is 2. The horizon agent manifold M_α is the two-dimensional triangular (hexagonal-packing) surface on which the confirmation frontier lives. A triangular lattice has two orientations of triangular face — upward and downward — and the two orientations enclose *distinct* bits with opposite circulation; this is the intrinsic two-fold structure of the flat surface itself, requiring no thickness. On any finite patch the two orientations occur in exactly equal number (a patch with 49 up-faces has 49 down-faces, and the equality is exact for every patch, being the standard bipartition of the triangular tiling). The bit capacity of the horizon is therefore the sum of the up-face bits and the down-face bits, exactly twice the single-orientation count: the factor of two is 2, derived, from the clean 1 : 1 up/down face pairing of the two-dimensional frontier.

The extent overhead $\epsilon^{(k)}$ is 0. The minimum region that can host a level- k closure has meta-extent $L_{\text{sub}}^{(k)} = 3L_p^{(k)} + \epsilon^{(k)}$, with $\epsilon^{(k)}$ the nudge required to seat the ideal closure onto the lattice. The closure is the regular tetrahedron drawn from four alternating cube corners — the stella-octangula A-sublattice cell — whose six edge-lengths-squared are all equal to the FCC nearest-neighbor value 2, and which has integer vertices at every integer scaling. The ideal tetrahedral closure therefore lands exactly on the FCC lattice at every scale, requiring no nudge: $\epsilon^{(k)} = 0$ identically, and $L_{\text{sub}}^{(k)} = 3L_p^{(k)}$ exactly. This also settles the $C(4, 2) = 6$ combinatorial-bound refinement that Volume 1 (Paper 4) carried forward: the six pair-distances of the closure cell are the six equal FCC edges, with no residual.

6. The Equivalence Principle as a Theorem of Depth

An equivalence-principle violation is, operationally, whether bodies of different composition fall differently, and so depends on the composition-depth difference between test bodies. The per-level mismatch $1/\sqrt{2} < 1$ compounds, suppressing the violation as $(1/\sqrt{2})^N$ where N is the depth at which the bodies' structures diverge. Ordinary matter differs only in shallow layers; N is large; the violation is unobservable; the equivalence principle holds. Einstein assumed the equivalence principle; the substrate derives it as a depth-limit. The sign (suppression) is derived; the magnitude is a bound.

7. Dynamic Dark Energy: the Forced Crossing

Dark energy is the relation-channel residual on the now-surface (Volume 1, p. 131). The now-surface advances every flop under $\hat{R}$; a residual on a perpetually advancing surface cannot hold constant density. Dark energy is therefore dynamic by necessity, and the standard-model value $w = -1$ is structurally excluded.

Early (a young, shallow universe), the now-surface advance dominates and the density grows: $w < -1$, phantom-like. Late (an old, deep universe), advance weakens while dilution and the depth-weight win: $w > -1$, quintessence-like. The equation of state should therefore cross -1 , from below to above, with the density peaking at intermediate redshift. This crossing distinguishes the substrate from the standard model, which forbids any crossing, and it qualitatively matches recent survey indications of evolving dark energy (DESI Collaboration 2025a) — w_0 above -1 today, a negative running, and a density peak near redshift one-half.

The status of this crossing must be stated carefully, and more cautiously than earlier drafts did. The *direction* of each limit is derived: a residual on a perpetually advancing surface cannot hold constant density, so $w = -1$ is excluded, and the young-universe (advance-dominated) and old-universe (dilution-and-depth-dominated) regimes have opposite signs of $w + 1$. That the equation of state therefore *crosses* -1 follows if the advance is dominant early and subdominant late — that is, if the two effects genuinely cross over. A forward simulation of the residual-density dynamics does not, however, robustly reproduce the crossing from generic inputs: depending on how the surface injection and its volume dilution are balanced, the model can remain phantom throughout or quintessence throughout, with the crossing appearing only for an intermediate balance. The crossing is therefore *not* parameter-free as an outcome, even though the two limiting signs are derived; whether the advance-dilution competition actually crosses over in our universe depends on the depth-accumulation history $k(a)$ and the injection normalization, which are not pinned here. The honest claim is thus weaker than a forced crossing: the mechanism *permits and motivates* a $w = -1$ crossing of the observed sign, and excludes the cosmological constant, but a forward simulation reproduces the crossing only for a range of the unpinned inputs rather than universally. The reproducibility script (*eos_{sim}*) records both the derived limiting signs and the input-dependence of the crossing. The exclusion of $w = -1$ is the robust prediction; the crossing itself is motivated but input-dependent.

Concordance with DESI DR2 (structural, not a derivation). Three features of the framework’s dark sector find qualitative echoes in the second data release of the Dark Energy Spectroscopic Instrument, and the concordance is recorded here at the level the framework earns — structural, not a fit. First, the framework excludes a cosmological constant and requires dynamical dark energy; DESI DR2 reports a preference for a time-evolving equation of state over Λ CDM in the quadrant $w_0 > -1$, $w_a < 0$, at up to the few- σ level depending on the supernova compilation (DESI Collaboration 2025a; Lodha et al. 2025). Second, the framework’s young-then-old sign structure places the equation of state in the phantom regime early and the quintessence regime late; DESI’s model-agnostic reconstructions favor exactly this phantom-to-quintessence ordering with a crossing near redshift 0.4, and report it to be robust to the choice of parametrization (Lodha et al. 2025). Third, and most specifically, the framework treats dark energy and dark matter not as independent components but as two channels of one substrate — the relation channel and the bit channel of $\bar{B} = 3\bar{\alpha}$ — which admits exchange between them; a growing line of DESI DR2 analysis reconstructs precisely such a dark-sector coupling, with the reconstructed interaction changing sign over cosmic history (energy passing from dark matter to dark energy early, reversing late) and favored over Λ CDM with strong Bayesian evidence in some data combinations (Li & Zhang 2025). This last feature is notable because the channel-exchange structure was fixed by the founding primitive, not chosen to match observation. One quantitative contact is worth recording alongside the qualitative ones, at the same arm’s length: DESI’s single best-measured,

parametrization-independent value is the pivot equation of state, $w(z \simeq 0.5) = -1.024 \pm 0.043$, and a forward run of the sector built on the framework's own formation and exhaustion relations (the reproducibility script *dark_{sector}*) returns $w \approx -1.02$ at that redshift without tuning to it — a coincidence at one point, well within the measurement error, and explicitly not a fit of the full curve. It is offered as a single un-tuned contact, not as a reproduction of the equation-of-state history, which the framework does not yet derive. None of this is offered as confirmation, and three caveats are stated plainly. First, these three features — dynamical dark energy, a phantom-to-quintessence crossing, and a sign-reversing dark-sector exchange — are *generic* to the broad class of beyond- Λ CDM models (quintessence, phantom, and interacting-dark-energy scenarios all produce some combination of them); the framework's concordance with them does not by itself discriminate it from that class, and would do so only once it derives the quantitative $w(z)$, which it does not yet. Second, the observational signal is itself contested and dataset-dependent: the sign-reversing coupling of Li & Zhang (2025) is favored at around the 2σ level for some data combinations (CMB + DESI + DESY5, CMB + DESI + Union3) but is consistent with no interaction for others (CMB + DESI, CMB + DESI + PantheonPlus), and separate analyses question whether the evolving-dark-energy signal is significant at all. Third, and to make the framework's exposure explicit rather than hedged: the sharp falsifier of the sector is a cosmological constant. If dark energy is observed to be constant — $w = -1$ within errors, with no dynamical evolution — then the framework's core dark-sector claim, that a residual on a perpetually advancing surface cannot hold constant density, is falsified. The framework is not immunized against this outcome; it is staked against it. What is recorded, then, is a structural concordance held at arm's length: the framework's qualitative dark-sector predictions match three independently reported trends in current data, those trends are generic and contested rather than decisive, the quantitative derivation is the sector's principal open problem, and the framework carries a definite falsifier should the cosmological constant prove correct after all.

8. The Single Integer

The framework's only empirical input is the universe's hierarchical depth k (Volume 1, p. 132: one geometric primitive, one axiom, one empirical question). Volume 1 (p. 131) pins $k \approx 79$ from the cosmological-constant ratio $\Lambda/\rho_p \approx 10^{-123}$ under the per-level scaling. This anchor — the vacuum-energy hierarchy — is independent of the equivalence-principle and dark-energy channels, and the same $k \approx 79$ is consistent with both: $(1/\sqrt{2})^{79} \approx 10^{-12}$, near the bound set by the MICROSCOPE equivalence-principle test (Touboul et al., 2022); and a deep universe gives the gentle equation-of-state evolution observed. The material universe reduces to $\sqrt{2}$, the Principle of Generative Economy, and one integer $k \approx 79$. This value is order-of-magnitude and rests on Volume 1's steady-state assumption (asserted, not derived); the claim is a triple-consistency at $k \approx 79$, not a precision determination.

9. Where It Sits

The now-surface results of §§ 1–8 were derived from the confirmation frontier as a structural object, without reference to the gravitational literature, so that the literature could be set against them. This section locates the paper among the programs in emergent and thermodynamic gravity it most resembles, and marks where it diverges. Placement is by coordinate; each program is taken as its authors state it; agreement is a coordinate reached after the derivation, not support for it.

The paper sits squarely within the thermodynamic and entropic-gravity program and forks from it at one definite point: what supplies the entropy. Jacobson derived the Einstein field equations as an equation of state, demanding the Clausius relation $\delta Q = T \delta S$ hold across every local causal horizon and recovering

the field equations from the area–entropy proportionality (Jacobson 1995); Verlinde read gravity itself as an entropic force, with the metric and the Newtonian potential emerging from an entropy gradient on a holographic screen (Verlinde 2011). The now-surface reproduces the skeleton of both — the field equations arise by the Jacobson construction on a derived area law and a re-derived Unruh-form temperature (§ 3), and the Poisson equation appears as the continuum shadow with the resolution deficit playing the potential (§ 3) — and diverges by supplying the microstructure the program leaves unspecified. Where Jacobson’s horizon and Verlinde’s screen are posited surfaces on which entropy is counted, the now-surface is a derived two-dimensional agent manifold whose confirmation flux carries the entropy (§ 2), so the screen is not an assumed boundary but the structural frontier Volume 1 already forced. The fork is therefore not over the thermodynamic route to the field equations, which the framework takes, but over whether the entropy-bearing surface is posited or derived.

The horizon thermodynamics the paper re-derives places it next to the Bekenstein–Hawking and Unruh results, again as correspondence rather than adoption. The area law sits on the horizon non-conditionally because the horizon is a two-dimensional agent manifold (§ 2), and the confirmation-frontier temperature is recovered in Unruh form with the horizon’s radiation following from the perspective cut (§ 4). These are landing points reached after the derivation: the framework does not assume the Bekenstein entropy or the Unruh temperature and then build on them, but re-derives their form from the now-surface and notes the agreement. The area–coupling factor of two is derived ($= 2$, § 5.5), so the correspondence is claimed with that coefficient fixed.

Two empirical contacts locate the paper against observation rather than against a rival theory, and both are forward-looking tests rather than confirmations claimed. The equivalence principle is derived as a depth-suppression theorem — a composition-dependent violation suppressed as $(1/\sqrt{2})^N$ in the depth at which test bodies diverge (§ 6) — which meets the null results of high-precision equivalence-principle tests as a sign-and-bound prediction, the sign derived and the magnitude bounded (Touboul et al. 2022). The forced crossing of the cosmological-constant value reads as dynamic dark energy (§ 7), which meets the evolving-equation-of-state signal reported by recent baryon-acoustic-oscillation surveys as a structural prediction with its numerical parameters left open (DESI Collaboration 2025). Both are recorded as coordinates with tests attached, not as vindications.

The placement is owned. The paper lands as an emergent-gravity account in the thermodynamic–entropic family that supplies a derived microstructure — the now-surface and its confirmation flux — where the established constructions posit a screen, and that meets the equivalence principle and dark-energy observations as derived predictions rather than fitted inputs. The framework does not argue this is the correct coordinate among emergent-gravity programs; it records that the structural object it derived falls here when set among them, and that the placement rests on the derivations of §§ 1–8 and not on a demonstration that the neighbouring programs are mistaken.

10. Summary and Placement

The now-surface, taken purely as a structural object, derives the material universe: the black hole as object, its Hawking emission, the metric in form, the gravitational coupling as one triadic alphabet, dark energy as a forced crossing of the cosmological-constant value, and the whole reducing to a single integer. This paper opens Volume 2 by establishing that the active confirmation frontier is physically load-bearing on the material side before any property of observers is introduced. The perspective boundary that appears at the black-hole horizon is the structural object on which the volume’s later analysis of bounded observers and graded subjecthood is built.

Paper 2 takes the now-surface as established here and inverts the standpoint: where this paper reads the confirmation frontier from outside, as material structure, Paper 2 reads it from within, developing the confirmation relation as the seat of bounded perspective and subjecthood as graded binding depth. The perspective boundary derived at the black-hole horizon is the structural hinge on which that inversion turns.

Open items carried forward: the sequential-sampling realization of $|S_3| = 6$; the numerical equation-of-state parameters from $k(a)$; and a formal, non-steady-state derivation of $k \approx 79$. The area-coupling factor of two ($= 2$), the geometric extent overhead $\epsilon^{(k)}$ ($= 0$), and the depth-coupling residual δ_k ($= 1$, forced by the form-invariance fixed point) are derived in § 5.4–5.5. The Hawking temperature’s absolute scale inherits the coupling identity but introduces no separate open problem.

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All theoretical commitments, the choice of axioms, the Principle of Generative Economy, the now-surface construction, and the framework’s overall architecture are the author’s; the AI systems’ role was to teach, formalize, critique, verify, and document.

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